

Evaporation

Pre-Lab Plan
Unit Operations Lab 1
January 23, 2020

Group 1, Thursday, PM section

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1. Experimental Objective

The objective of this lab is to concentrate a solution which consists of a nonvolatile solute and a volatile solvent. Evaporation, a heat transfer operation, is used to produce a concentrated solution or thick liquor and some important properties of evaporating liquids are concentration, foaming, temperature sensitivity, scale on heating surfaces, and materials of construction.

2. Experimental

2.1 Apparatus

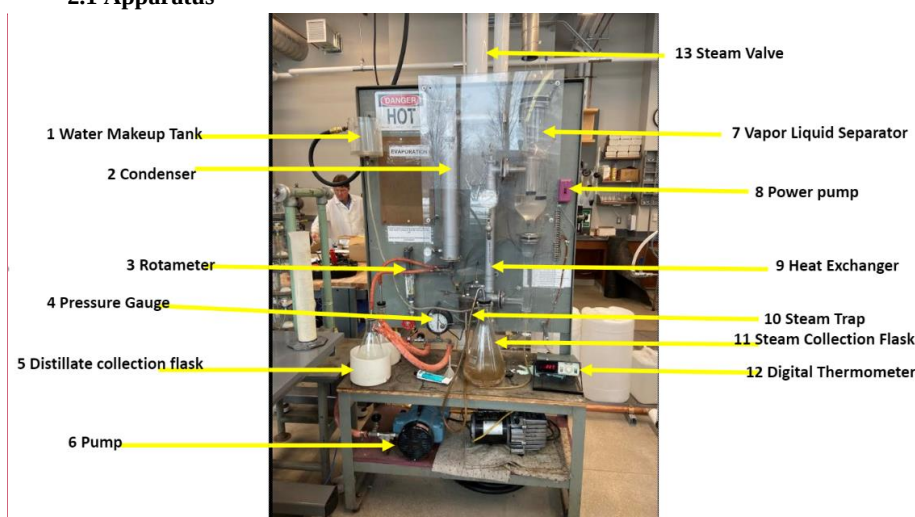


Figure 1: Main Apparatus



Figure 2: Steam Valve

Commented [DL2]: 3/5

Commented [AZ3]: Actually you won't be concentrating the solution but performing the experiments at a constant concentration (that is what the makeup water is for).

Commented [DL4]: What is the nonvolatile solute and the volatile solvent particularly in this lab?

Commented [AZ5]: Provide more details on the objective and the apparatus (single-effect evaporator).

Commented [DL6]: What is going to be observed during the experiment?

Commented [DL7]: 3.5/5
A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

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Table 1: Apparatus Components

Number	Name	Description
1	Water Makeup Tank	Holds water
2	Condenser	Convert evaporated vapor to liquid
3	Rotameter	Measures the flow rate of condensed liquid
4	Pressure Gauge	Measures pressure of system
5	Distillate Collection Flask	Collects the distillate from the vessel
6	Vacuum Pump	Pressurizes the system
7	Vapor Liquid Separator	Holds the Sugar mixture for evaporating
8	Power Pump	Switch for power of the vacuum pump
9	Heat Exchanger	Heats makeup water using steam
10	Steam Trap	Controls steam
11	Steam Collection Flask	Collects resulting steam from vessel
12	Digital Thermometer Reader	Measures Temperature at steady state
13	Steam Valve	Opens and closes steam flow

Commented [DL9]: Unit and the maximum capacity.

Commented [DL10]: Unit. And other gauges should be mentioned.

Commented [DL11]: Create vacuum in the apparatus.

Commented [DL12]: Pump switch.

2.2 Materials and Supplies

Table 2: Materials and Supplies

Materials/supplies	Usage
Cooling Water	Used in the condenser
Electricity	Used to run the pump
Sugar and water	Used to mix the sugar solution
Heat Resistant Gloves	Safety precaution to protect from burns
Nitrile Gloves	Safety precaution to protect from skin contact
Steam	Used in the heat exchanger

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Scoopula, scale, vacuum grease, timer and funnel are needed.

2.3 Experimental Plan

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Experimental Procedure:

1. Drain contents of the separator through a valve at the bottom of the separator.
2. Prepare a 10 wt% sugar solution and fill the separator to the level line. Allow some of the solution to flow into the tube used to drain the separator.
3. Put pure water in the makeup tank- adjust the flow of makeup water until it is at a constant level. Start the flow of steam into the system.
4. Start the cooling water to the steam trap. Collect the steam condensate in a graduated cylinder.

Commented [DL15]: How much?

Commented [DL16]: Which one should be the first?

5. Turn on the vacuum pump and adjust the bypass valve to obtain a desired pressure in the separator.
6. Adjust the vacuum so that a steady flow is achieved.
7. Collect the vapor product (after it has condensed). Be sure to add enough makeup water to maintain the level at a steady state.
8. Wait for the system to reach a steady state as indicated by the temperature readings for dial setting 1 and 2.
9. After reaching a steady state, note the level of the steam condensate and collected vapor. Now, wait for 15 minutes and read these levels again.
10. Record the flow rate of the cooling water to the steam trap, and the inlet and outlet temperature.
11. Now, change the pump to a different vacuum level, for a total of three vacuum settings.

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Commented [DL18]: Good.

Table 3: Independent Variables in Evaporation Experiment

Independent Variables	Notes
Pressure	For three trials: 5, 10, 15 psi will be used

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Table 4: Dependent Variables in Evaporation Experiment

Dependent Variables	Notes
Cooling Water Flow rate	Record the flow rate of water after 15 minutes of each trial
Inlet Temperature	Record the inlet temperature after 15 minutes of each trial
Outlet Temperature	Record the inlet temperature after 15 minutes of each trial
Distillate	Collection of water out of vessel
Steam Distillate	Collection of condensed steam

There will be three trials total with pressures at 5, 10, and 15 inches of mercury. The cooling water flowrate will be recorded for each trial. In each trial we will time the amount of product and steam obtained in minutes as well as the volumes in millimeters. The inlet and outlet temperatures will be read in Celsius.

Commented [DL20]: Not consistent with what above.

Commented [DL21]: Be careful about the unit.

Project Safety Evaluation

Project Title: Evaporation

Date: 1/23/2020

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	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Electrical outlets are located right behind apparatus and for pump; apparatus has water circulating constantly- interaction between water and electricity can cause electrocution if in contact.
Potential mechanical hazards?	Yes	If yes, identify conditions: If in contact with pump inlet, there is potential for injury.
Potential pressure hazards?	Yes	If yes, identify conditions: Controlling pressure manually through pump, if not controlled, can cause over pressurized system.
Potential temperature hazards?	Yes	If yes, identify conditions: High temperature steam discharge in flask for collection
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Heat resistant gloves for flask handling
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water: CAS- 7732-18-5 Sugar: CAS- 57-50-1
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	100°C	The heat exchanger and steam water along with the apparatus holding the sugar solution when heated can cause burns when contacted with bare hands
Pressure (psia)	5-15 psi	Inside the vessel
Safety Controls	yes/no	If yes, Provide description
Thermometer, installat	Yes	Steam inlet from steam valve to regulate the amount of steam coming in system Thermometer on the heat exchanger to make sure safe temperature of heat Pump plug to regulate amount pressure of system- can be quickly turned on/off

Commented [DL23]: Thermocouples.

Commented [DL24]: About 120 degree Celsius.

I have read relevant background material for the Unit Operations Laboratory entitled: “ Evaporation ” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Advith Vashist</u> signature of team member	<u>1/23/20</u> date
<u>Dulce Colindres</u> signature of team member	<u>1/23/20</u> date
<u>Kathya Chavez</u> signature of team member	<u>1/23/20</u> date

<u>Marium Murtaza</u>	<u>1/23/20</u>
signature of team member	date
<u>Yaser Khalid</u>	<u>1/23/20</u>
signature of team member	date